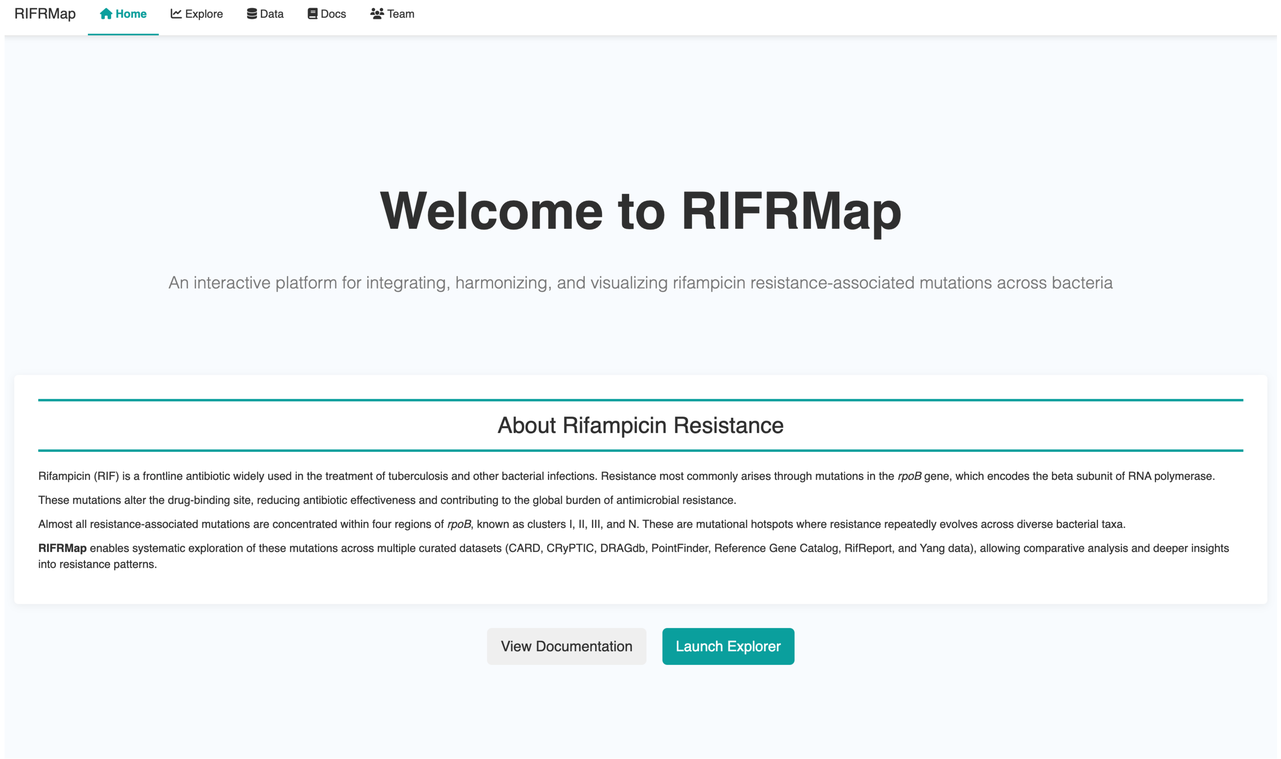
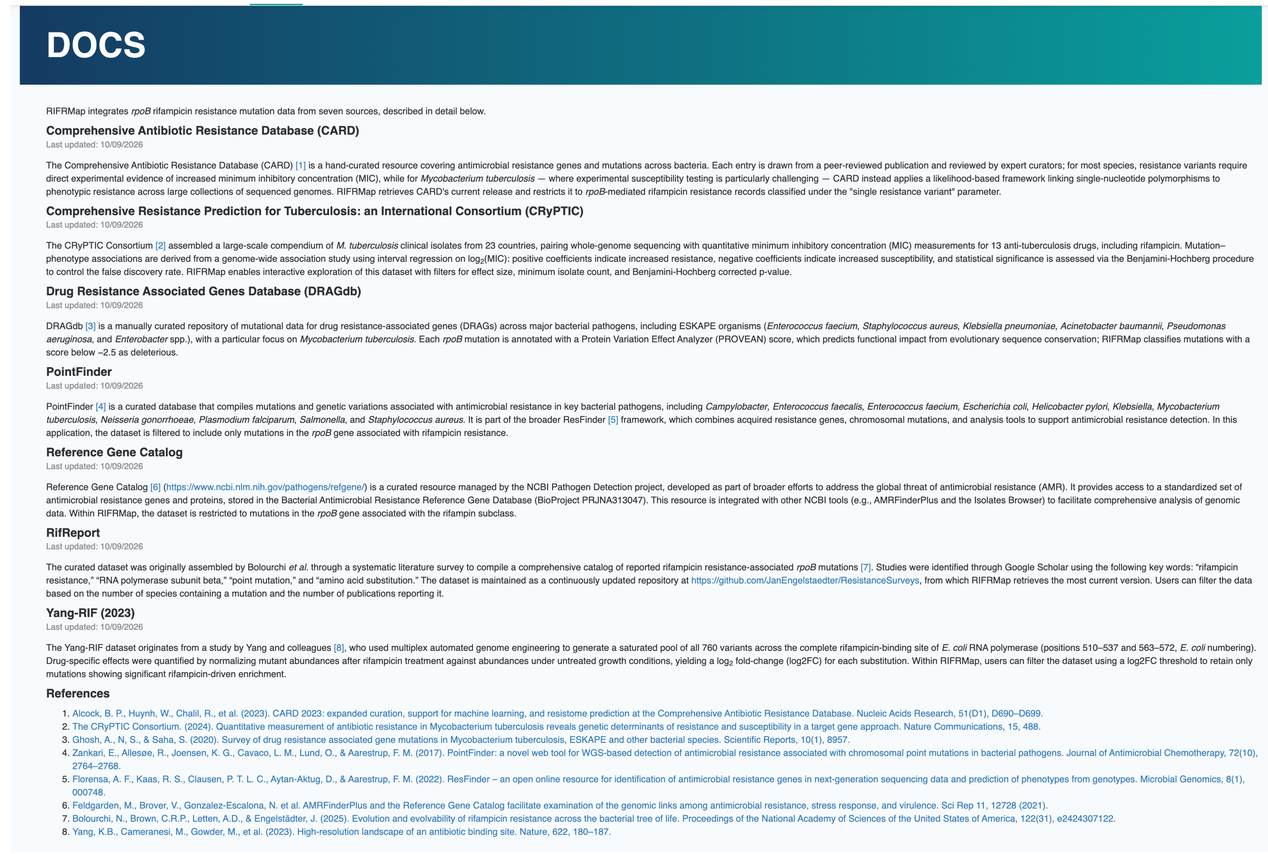
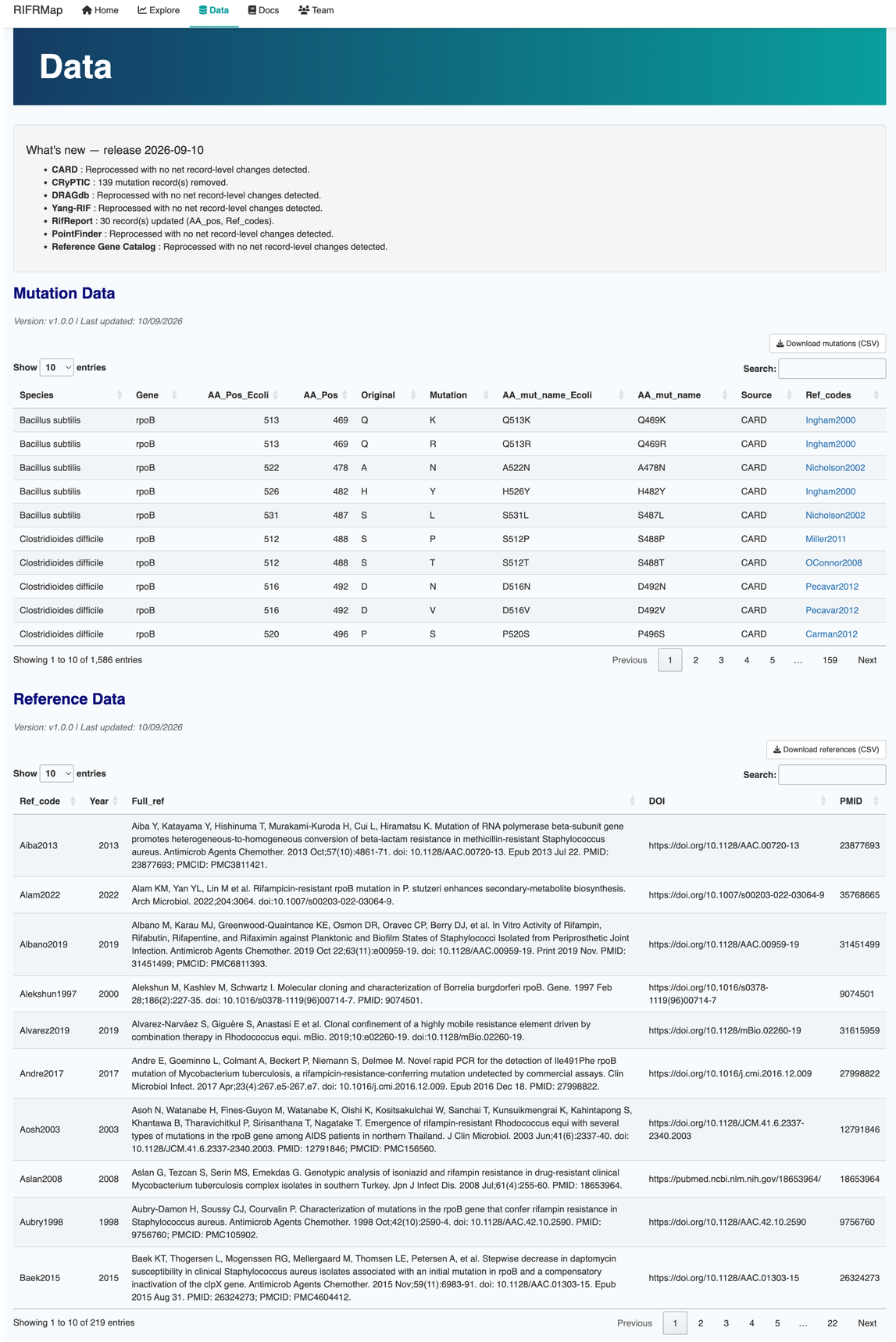
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Welcome to RIFRMap
An interactive platform for integrating, harmonizing, and visualizing rifampicin resistance-associated mutations across bacteria
About Rifampicin Resistance
Rifampicin (RIF) is a frontline antibiotic widely used in the treatment of tuberculosis and other bacterial infections. Resistance most commonly arises through mutations in the rpoB gene, which encodes the beta subunit of RNA polymerase.
These mutations alter the drug-binding site, reducing antibiotic effectiveness and contributing to the global burden of antimicrobial resistance.
Almost all resistance-associated mutations are concentrated within four regions of rpoB, known as clusters I, II, III, and N. These are mutational hotspots where resistance repeatedly evolves across diverse bacterial taxa.
RIFRMap enables systematic exploration of these mutations across multiple curated datasets (CARD, CRyPTIC, DRAGdb, PointFinder, Reference Gene Catalog, RifReport, and Yang data), allowing comparative analysis and deeper insights into resistance patterns.
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RIFRMap integrates rpoB rifampicin resistance mutation data from seven sources, described in detail below.
Comprehensive Antibiotic Resistance Database (CARD)
Last updated: 10/09/2026
The Comprehensive Antibiotic Resistance Database (CARD) [1] is a hand-curated resource covering antimicrobial resistance genes and mutations across bacteria. Each entry is drawn from a peer-reviewed publication and reviewed by expert curators; for most species, resistance variants require direct experimental evidence of increased minimum inhibitory concentration (MIC), while for Mycobacterium tuberculosis — where experimental susceptibility testing is particularly challenging — CARD instead applies a likelihood-based framework linking single-nucleotide polymorphisms to phenotypic resistance across large collections of sequenced genomes. RIFRMap retrieves CARD's current release and restricts it to rpoB-mediated rifampicin resistance records classified under the "single resistance variant" parameter.
Comprehensive Resistance Prediction for Tuberculosis: an International Consortium (CRyPTIC)
Last updated: 10/09/2026
The CRyPTIC Consortium [2] assembled a large-scale compendium of M. tuberculosis clinical isolates from 23 countries, pairing whole-genome sequencing with quantitative minimum inhibitory concentration (MIC) measurements for 13 anti-tuberculosis drugs, including rifampicin. Mutation–phenotype associations are derived from a genome-wide association study using interval regression on log2(MIC): positive coefficients indicate increased resistance, negative coefficients indicate increased susceptibility, and statistical significance is assessed via the Benjamini-Hochberg procedure to control the false discovery rate. RIFRMap enables interactive exploration of this dataset with filters for effect size, minimum isolate count, and Benjamini-Hochberg corrected p-value.
Drug Resistance Associated Genes Database (DRAGdb)
Last updated: 10/09/2026
DRAGdb [3] is a manually curated repository of mutational data for drug resistance-associated genes (DRAGs) across major bacterial pathogens, including ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter spp.), with a particular focus on Mycobacterium tuberculosis. Each rpoB mutation is annotated with a Protein Variation Effect Analyzer (PROVEAN) score, which predicts functional impact from evolutionary sequence conservation; RIFRMap classifies mutations with a score below −2.5 as deleterious.
PointFinder
Last updated: 10/09/2026
PointFinder [4] is a curated database that compiles mutations and genetic variations associated with antimicrobial resistance in key bacterial pathogens, including Campylobacter, Enterococcus faecalis, Enterococcus faecium, Escherichia coli, Helicobacter pylori, Klebsiella, Mycobacterium tuberculosis, Neisseria gonorrhoeae, Plasmodium falciparum, Salmonella, and Staphylococcus aureus. It is part of the broader ResFinder [5] framework, which combines acquired resistance genes, chromosomal mutations, and analysis tools to support antimicrobial resistance detection. In this application, the dataset is filtered to include only mutations in the rpoB gene associated with rifampicin resistance.
Reference Gene Catalog
Last updated: 10/09/2026
Reference Gene Catalog [6] (https://www.ncbi.nlm.nih.gov/pathogens/refgene/) is a curated resource managed by the NCBI Pathogen Detection project, developed as part of broader efforts to address the global threat of antimicrobial resistance (AMR). It provides access to a standardized set of antimicrobial resistance genes and proteins, stored in the Bacterial Antimicrobial Resistance Reference Gene Database (BioProject PRJNA313047). This resource is integrated with other NCBI tools (e.g., AMRFinderPlus and the Isolates Browser) to facilitate comprehensive analysis of genomic data. Within RIFRMap, the dataset is restricted to mutations in the rpoB gene associated with the rifampin subclass.
RifReport
Last updated: 10/09/2026
The curated dataset was originally assembled by Bolourchi et al. through a systematic literature survey to compile a comprehensive catalog of reported rifampicin resistance-associated rpoB mutations [7]. Studies were identified through Google Scholar using the following key words: "rifampicin resistance," "RNA polymerase subunit beta," "point mutation," and "amino acid substitution." The dataset is maintained as a continuously updated repository at https://github.com/JanEngelstaedter/ResistanceSurveys, from which RIFRMap retrieves the most current version. Users can filter the data based on the number of species containing a mutation and the number of publications reporting it.
Yang-RIF (2023)
Last updated: 10/09/2026
The Yang-RIF dataset originates from a study by Yang and colleagues [8], who used multiplex automated genome engineering to generate a saturated pool of all 760 variants across the complete rifampicin-binding site of E. coli RNA polymerase (positions 510–537 and 563–572, E. coli numbering). Drug-specific effects were quantified by normalizing mutant abundances after rifampicin treatment against abundances under untreated growth conditions, yielding a log2 fold-change (log2FC) for each substitution. Within RIFRMap, users can filter the dataset using a log2FC threshold to retain only mutations showing significant rifampicin-driven enrichment.
References
1. Alcock, B. P., Huynh, W., Chalil, R., et al. (2023). CARD 2023: expanded curation, support for machine learning, and resistome prediction at the Comprehensive Antibiotic Resistance Database. Nucleic Acids Research, 51(D1), D690–D699.
2. The CRyPTIC Consortium. (2024). Quantitative measurement of antibiotic resistance in Mycobacterium tuberculosis reveals genetic determinants of resistance and susceptibility in a target gene approach. Nature Communications, 15, 488.
3. Ghosh, A., N, S., & Saha, S. (2020). Survey of drug resistance associated gene mutations in Mycobacterium tuberculosis, ESKAPE and other bacterial species. Scientific Reports, 10(1), 8957.
4. Zankari, E., Allesøe, R., Joensen, K. G., Cavaco, L. M., Lund, O., & Aarestrup, F. M. (2017). PointFinder: a novel web tool for WGS-based detection of antimicrobial resistance associated with chromosomal point mutations in bacterial pathogens. Journal of Antimicrobial Chemotherapy, 72(10), 2764–2768.
5. Florensa, A. F., Kaas, R. S., Clausen, P. T. L. C., Aytan-Aktug, D., & Aarestrup, F. M. (2022). ResFinder – an open online resource for identification of antimicrobial resistance genes in next-generation sequencing data and prediction of phenotypes from genotypes. Microbial Genomics, 8(1), 000748.
6. Feldgarden, M., Brover, V., Gonzalez-Escalona, N. et al. AMRFinderPlus and the Reference Gene Catalog facilitate examination of the genomic links among antimicrobial resistance, stress response, and virulence. Sci Rep 11, 12728 (2021).
7. Bolourchi, N., Brown, C.R.P., Letten, A.D., & Engelstädter, J. (2025). Evolution and evolvability of rifampicin resistance across the bacterial tree of life. Proceedings of the National Academy of Sciences of the United States of America, 122(31), e2424307122.
8. Yang, K.B., Cameranesi, M., Gowder, M., et al. (2023). High-resolution landscape of an antibiotic binding site. Nature, 622, 180–187.

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What's new — release 2026-09-10
CARD : Reprocessed with no net record-level changes detected.
CRyPTIC : 139 mutation record(s) removed.
DRAGdb : Reprocessed with no net record-level changes detected.
Yang-RIF : Reprocessed with no net record-level changes detected.
RifReport : 30 record(s) updated (AA_pos, Ref_codes).
PointFinder : Reprocessed with no net record-level changes detected.
Reference Gene Catalog : Reprocessed with no net record-level changes detected.
Mutation Data
Version: v1.0.0 | Last updated: 10/09/2026
Download mutations (CSV)
Show 10 entries
Search:
Species | Gene | AA_Pos_Ecoli | AA_Pos | Original | Mutation | AA_mut_name_Ecoli | AA_mut_name | Source | Ref_codes
Bacillus subtilis | rpoB | 513 | 469 | Q | K | Q513K | Q469K | CARD | Ingham2000
Bacillus subtilis | rpoB | 513 | 469 | Q | R | Q513R | Q469R | CARD | Ingham2000
Bacillus subtilis | rpoB | 522 | 478 | A | N | A522N | A478N | CARD | Nicholson2002
Bacillus subtilis | rpoB | 526 | 482 | H | Y | H526Y | H482Y | CARD | Ingham2000
Bacillus subtilis | rpoB | 531 | 487 | S | L | S531L | S487L | CARD | Nicholson2002
Clostridioides difficile | rpoB | 512 | 488 | S | P | S512P | S488P | CARD | Miller2011
Clostridioides difficile | rpoB | 512 | 488 | S | T | S512T | S488T | CARD | OConnor2008
Clostridioides difficile | rpoB | 516 | 492 | D | N | D516N | D492N | CARD | Pecavar2012
Clostridioides difficile | rpoB | 516 | 492 | D | V | D516V | D492V | CARD | Pecavar2012
Clostridioides difficile | rpoB | 520 | 496 | P | S | P520S | P496S | CARD | Carman2012
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Reference Data
Version: v1.0.0 | Last updated: 10/09/2026
Download references (CSV)
Show 10 entries
Search:
Ref_code | Year | Full_ref | DOI | PMID
Aiba2013 | 2013 | Aiba Y, Katayama Y, Hishinuma T, Murakami-Kuroda H, Cui L, Hiramatsu K. Mutation of RNA polymerase beta-subunit gene promotes heterogeneous-to-homogeneous conversion of beta-lactam resistance in methicillin-resistant Staphylococcus aureus. Antimicrob Agents Chemother. 2013 Oct;57(10):4861-71. doi: 10.1128/AAC.00720-13. Epub 2013 Jul 22. PMID: 23877693; PMCID: PMC3811421. | https://doi.org/10.1128/AAC.00720-13 | 23877693
Alam2022 | 2022 | Alam KM, Yan YL, Lin M et al. Rifampicin-resistant rpoB mutation in P. stutzeri enhances secondary-metabolite biosynthesis. Arch Microbiol. 2022;204:3064. doi:10.1007/s00203-022-03064-9. | https://doi.org/10.1007/s00203-022-03064-9 | 35768665
Albano2019 | 2019 | Albano M, Karau MJ, Greenwood-Quaintance KE, Osmon DR, Oravec CP, Berry DJ, et al. In Vitro Activity of Rifampin, Rifabutin, Rifapentine, and Rifaximin against Planktonic and Biofilm States of Staphylococci Isolated from Periprosthetic Joint Infection. Antimicrob Agents Chemother. 2019 Oct 22;63(11):e00959-19. doi: 10.1128/AAC.00959-19. Print 2019 Nov. PMID: 31451499; PMCID: PMC6811393. | https://doi.org/10.1128/AAC.00959-19 | 31451499
Alekshun1997 | 2000 | Alekshun M, Kashlev M, Schwartz I. Molecular cloning and characterization of Borrelia burgdorferi rpoB. Gene. 1997 Feb 28;186(2):227-35. doi: 10.1016/s0378-1119(96)00714-7. PMID: 9074501. | https://doi.org/10.1016/s0378-1119(96)00714-7 | 9074501
Alvarez2019 | 2019 | Alvarez-Narváez S, Giguère S, Anastasi E et al. Clonal confinement of a highly mobile resistance element driven by combination therapy in Rhodococcus equi. mBio. 2019;10:e02260-19. doi:10.1128/mBio.02260-19. | https://doi.org/10.1128/mBio.02260-19 | 31615959
Andre2017 | 2017 | Andre E, Goeminne L, Colmant A, Beckert P, Niemann S, Delmee M. Novel rapid PCR for the detection of Ile491Phe rpoB mutation of Mycobacterium tuberculosis, a rifampicin-resistance-conferring mutation undetected by commercial assays. Clin Microbiol Infect. 2017 Apr;23(4):267.e5-267.e7. doi: 10.1016/j.cmi.2016.12.009. Epub 2016 Dec 18. PMID: 27998822. | https://doi.org/10.1016/j.cmi.2016.12.009 | 27998822
Aosh2003 | 2003 | Asoh N, Watanabe H, Fines-Guyon M, Watanabe K, Oishi K, Kositsakulchai W, Sanchai T, Kunsuikmengrai K, Kahintapong S, Khantawa B, Tharavichitkul P, Sirisanthana T, Nagatake T. Emergence of rifampin-resistant Rhodococcus equi with several types of mutations in the rpoB gene among AIDS patients in northern Thailand. J Clin Microbiol. 2003 Jun;41(6):2337-40. doi: 10.1128/JCM.41.6.2337-2340.2003. PMID: 12791846; PMCID: PMC156560. | https://doi.org/10.1128/JCM.41.6.2337-2340.2003 | 12791846
Aslan2008 | 2008 | Aslan G, Tezcan S, Serin MS, Emekdas G. Genotypic analysis of isoniazid and rifampin resistance in drug-resistant clinical Mycobacterium tuberculosis complex isolates in southern Turkey. Jpn J Infect Dis. 2008 Jul;61(4):255-60. PMID: 18653964. | https://pubmed.ncbi.nlm.nih.gov/18653964/ | 18653964
Aubry1998 | 1998 | Aubry-Damon H, Soussy CJ, Courvalin P. Characterization of mutations in the rpoB gene that confer rifampin resistance in Staphylococcus aureus. Antimicrob Agents Chemother. 1998 Oct;42(10):2590-4. doi: 10.1128/AAC.42.10.2590. PMID: 9756760; PMCID: PMC105902. | https://doi.org/10.1128/AAC.42.10.2590 | 9756760
Baek2015 | 2015 | Baek KT, Thogersen L, Mogenssen RG, Mellergaard M, Thomsen LE, Petersen A, et al. Stepwise decrease in daptomycin susceptibility in clinical Staphylococcus aureus isolates associated with an initial mutation in rpoB and a compensatory inactivation of the clpX gene. Antimicrob Agents Chemother. 2015 Nov;59(11):6983-91. doi: 10.1128/AAC.01303-15. Epub 2015 Aug 31. PMID: 26324273; PMCID: PMC4604412. | https://doi.org/10.1128/AAC.01303-15 | 26324273
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